

[Home](#) > [Media](#) > [Newsroom](#) >

Managing impact in Hoboken

March 17, 2021

Text in: [English](#)

The Umicore site in Hoboken is one of the world's largest precious metals refiners, offering recycling and refining services for precious metal bearing materials such as by-products from industry, electronic scrap and spent industrial and automotive catalysts. For several decades, the blood levels of children living close to the plant have been monitored twice per year by the authorities. In July 2020 the readings of the children living close to the Umicore recycling plant in Hoboken, Belgium, showed elevated levels of lead in their blood after multiple years of steady decreases, with historically low levels in 2019. Although the root cause investigation showed no major direct source of lead emissions in the plant, Umicore is aware that the risks associated with lead need to be well managed at all times. Various actions have been taken to reduce diffuse emissions and to ensure that the local community and industry can co-exist sustainably.

Lead in Umicore's process

Umicore processes complex materials from end-of-life products and intermediates from the mining and metallurgical industries. In its Hoboken recycling facility, it recovers 17 different metals, among which seven

[Menu](#) 

increasing scarcity of certain metals, making it a key enabler in the circular economy. Metals are at the heart of innovation and at the core of everyday life.

The Umicore site in Hoboken is one of the world's largest precious metals refiners, offering recycling and refining services for precious metal bearing materials such as by-products from industry, electronic scrap and spent industrial and automotive catalysts.

Industry and community, a long history of coexistence and monitoring

In 1887, the German companies Metallgesellschaft and Degussa founded the Société Générale Metallurgique in Hoboken along the banks of the river Scheldt to extract silver from lead ore concentrates and by-products. In the same time period the industrial development in the area took place and shipyards and other factories were built. More and more people moved to Hoboken and there was a need for additional housing. This led to the development of several housing quarters. In 1901, the first part of the Moretusburg residential area was constructed to provide housing for workers of the nearby industry on property owned by the family Moretus. The area was developed, without any kind of buffer zone, it grew historically. It is fair to say there was insufficient awareness of the impact of industry on a nearby residential area and the potentially adverse effects on the health of local communities. Construction happened in several phases over the 20th century. The houses closest to the plant were constructed in 1958.

However, the emissions of lead in the neighborhood caused severe pollution and blood samples in 1978 showed that children who lived in Moretusburg had very high levels of lead in blood. The concentration was higher as the children lived closer to the plant. From that moment on the [COOKIE SETTINGS](#) started a biannual monitoring program, while the company had

[Menu](#) 

the plant, the introduction of new technologies and better hygiene measures the situation improved dramatically. Finally, in 2006-2008, a cleaning of the neighborhood houses and yards completed a clean-up program of historical pollution. Meanwhile, emission norms were progressively getting stricter and stricter and were reduced to the levels of today.

Levels of lead in blood around the Hoboken plant

For several decades, the blood levels of children living close to the plant have been monitored twice per year by the authorities. In recent years, the values have been below the level of 5 micrograms of lead per deciliter of blood which is the reference value defined by the Center of Disease Control (USA).

At the end of 2015, an increase in emissions led to a surge in the lead-in blood values of children living close to the plant. The reason for this increase was a roof reparation at the lead refinery which resulted in the accidental release of lead dust. As a result, Umicore embarked on an ambitious and comprehensive investment program, including a new dust extraction system of the full lead refinery. The extraction cleaning is operational since the second half of 2018, after which we saw further improvement of the environmental results, as well as the lead in blood levels of children living close by.

In July 2020 the readings of the children living close to the Umicore recycling plant in Hoboken, Belgium, showed elevated levels of lead in their blood after multiple years of steady decreases, with historically low levels in 2019. This sudden rise in the readings for lead in blood came unexpectedly, as the emissions from the plant had consistently been well below the legal norm. The results of further tests made in October 2020 showed once again a clear reduction in the levels, with an average lead

below the reference value of 5µg/dl.

COOKIE SETTINGS

Root causes and operational response

When the results of the spring campaign were known, Umicore performed an in-depth analysis of the root causes and took measures to return to the positive trend of the previous years. After analysis it was concluded that external factors, such as exceptional weather conditions with high wind speeds and storms, combined with the increased exposure of the children, who spent several months at home due to the COVID-19 lockdown measures, were the root cause of the elevated readings.

Umicore initiated numerous measures to return to the positive trend of recent years. More intensive cleaning of roads and squares at the plant and in the neighborhood was initiated, both frequency and thoroughness of cleaning was intensified. New storage areas for raw materials are now fully covered. Because weather patterns have changed due to climate change, Umicore will take these extreme weather situations into account and will include them in its procedures. The weather barometer has been introduced, whereby the handling and movement of raw materials is limited or postponed depending on the weather conditions. New measuring techniques are being tested and applied. All potential sources of dust (including small ones) at the plant are investigated. The aim is to implement improvement measures and to simulate the influence that these sources can have on the neighborhood. This research focuses on the immediate reduction of diffuse emissions, as well as on the more fundamental adjustment of processes. Already in the second half of the year, this study led to improvement actions with measurable results. Investments to reduce dust nuisance are given priority (as in the case of fire prevention) over investments in new installations.

Umicore made multiple environmental improvements over the years, such as encapsulating the full lead refinery and investing € 25 million per annum to further increase eco-efficiency. With regard to fire prevention, several

COOKIE SETTINGS

been started or carried out to reinforce Umicore's fire and

[Menu](#) 

Sustainably minimizing lead impact by creating a green zone

Various actions have been taken to reduce diffuse emissions and to ensure that the local community and industry can co-exist sustainably. Together with the city of Antwerp, Umicore organizes activities for children outside the area during school holidays until a sustainable improvement in lead in blood values is established.

Although the root cause investigation showed no major direct source of lead emissions in the plant, Umicore is aware that the risks associated with lead need to be well managed at all times. In addition to the above measures, Umicore has offered to purchase the houses located closest to the plant on a voluntary basis in order to create a green zone and thereby increasing the distance between the residential area and the site. At the same time, Umicore will also create a green zone on its operational site, adjacent to the Moretusburg area. Concertation with the city council and the residents is essential and an open dialogue with all stakeholders is paramount to keep them fully informed of Umicore's activities, its environmental results and initiatives.

Adequate and effective emission control measures are required so that any adverse effects on the environment are minimized. Therefore, improvements will be pursued as our research teams continue to search for possible sources and improvements to ensure that our neighbors can live in an healthy and safe way.

More stories

[COOKIE SETTINGS](#)



Menu 

[Prebes Award](#)



[New business incubator as part of umicore's innovation strategy](#)



[Thermal imaging in the fight against COVID-19](#)



[Pioneering the sourcing and recycling of sustainable germanium](#)



[Umicore innovates with virtual trade show](#)



[Umicore fuel cell catalysts](#)



[Exemplary management of historical radioactive waste stored in Olen](#)



More news



Half Year Results 2025

PRESS RELEASE August 1, 2025



From Lab to Road

NEWS July 30, 2025



Advanced notice of Umicore's 2025 Half Year Results

PRESS RELEASE July 15, 2025



Menu 



Markets & products
Join us

About Umicore

Sustainability
Investor relations
Media

Innovation
Locations
Contact



© 2025 Umicore [Terms of use](#) [General terms & conditions](#) [Privacy and cookie notice](#)
[Supplier zone](#) [Integrity line](#)